

Levi Webb^{1,2}, Paul Grabowski², Jim Glosli², Matt Caplan¹, and Thomas Gomez³¹Illinois State University, Normal, IL 61761, USA²Lawrence Livermore National Laboratory, Livermore, CA 94550, USA³University of Colorado, Boulder, CO 80309, USA

Abstract

Many properties used to study and classify astronomical objects depend upon the opacities of their constituent plasmas. Spectral line shapes are essential for determining the temperature and density of observed stellar and compact objects and understanding and accurately modeling plasma opacities is crucial for computational stellar astrophysics. However, there are discrepancies between observed and calculated line shapes in the high energy density (HED) regime. We explore various plasma line shape models to ascertain potential sources of these discrepancies. Current opacity models use semi-empirical molecular dynamics (MD) to simulate local particle electric fields (microfields), possibly contributing to the observed discrepancies in line shape calculations. Employing physically motivated MD models at these conditions may rectify this by improving microfield calculations. We demonstrate the microfield capabilities of LLNL's MD code ddcMD for Stark broadening, in preparation for a future merge with the pre-existing line broadening program Xenomorph to enable more accurate line shape simulations.

Plasma

- Partially or fully ionized hot gas, often considered the fourth state of matter.
- Found in many places, including on Earth in lightning and ionosphere.
- High energy density (HED) plasmas are found in stars and compact objects, such as white dwarfs (WDs) and neutron stars (NSs). HED plasmas can behave as fluids or solids, depending on their conditions. For example, the cores of WDs solidify as the dead stars cool.
- Since plasmas make up stars and compact objects, understanding how these bodies behave depends heavily upon our knowledge of astrophysical plasmas.



Stellar Spectra

- The most abundant data from stars & compact objects come from their light.
- Many stellar properties—including luminosity (brightness), structure, lifespan, & temperature—are determined by opacity effects that govern how light interacts with stellar material at different densities.
- Astronomers extract these properties via stellar spectra, making it imperative that theoretical and computational models match these observations.

Fig. 1: An example of a stellar spectrum depicting intensity versus wavelength. These absorption line profiles contain valuable information about elemental abundances and other conditions within the star.

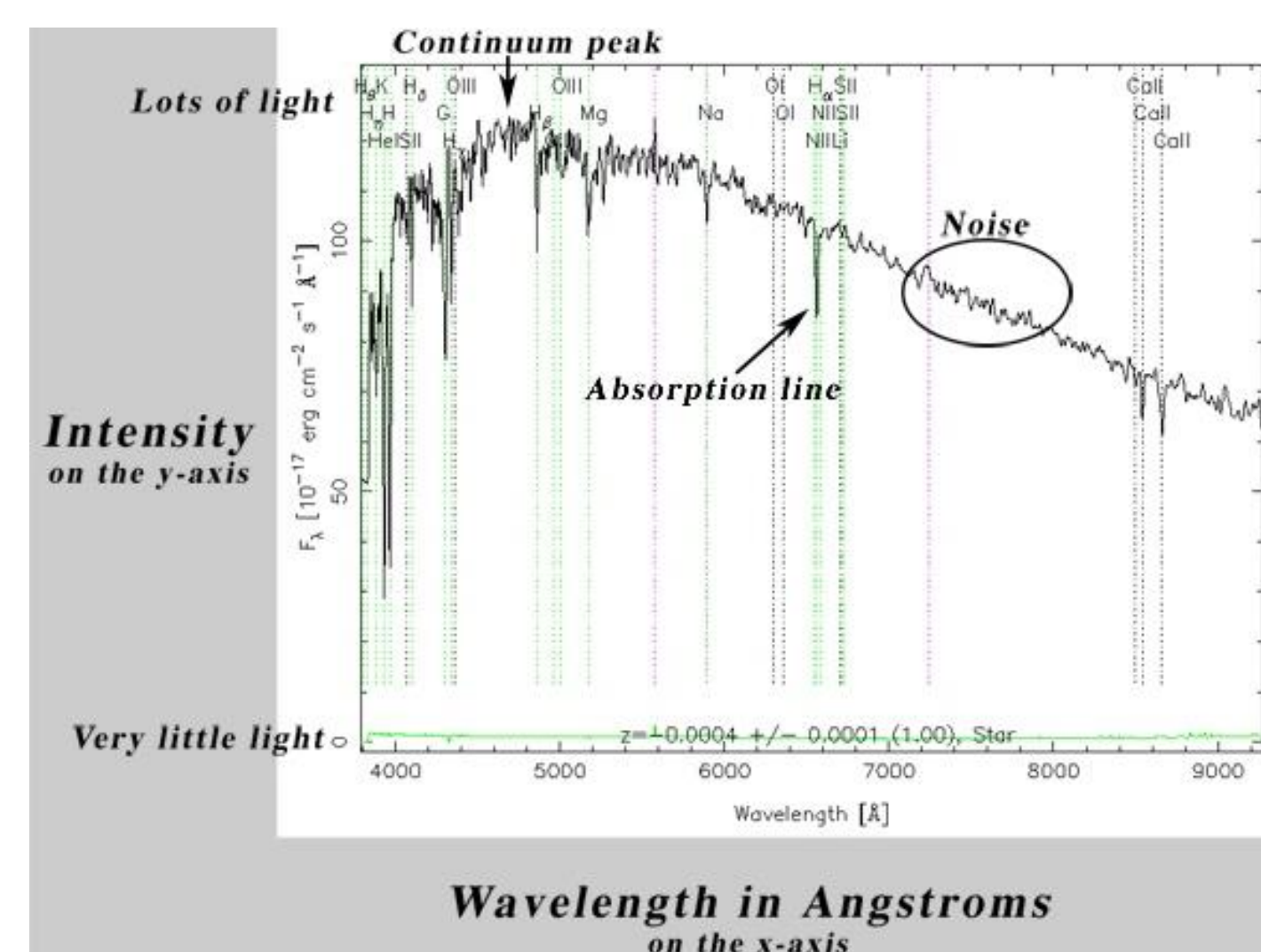


Image credit: SDSS

Opacity Line Shapes

- While absorption features appear as intensity dips in observed spectra, opacity line shapes show the inverse: opaque material blocks more light, creating peaks where intensity spectra show dips.
- One factor that influences line shapes is the presence of an electric field, which largely affects the opacity spectrum via processes like Stark broadening. It can also cause line merging and pressure ionization, both of which affect the continuum opacity.

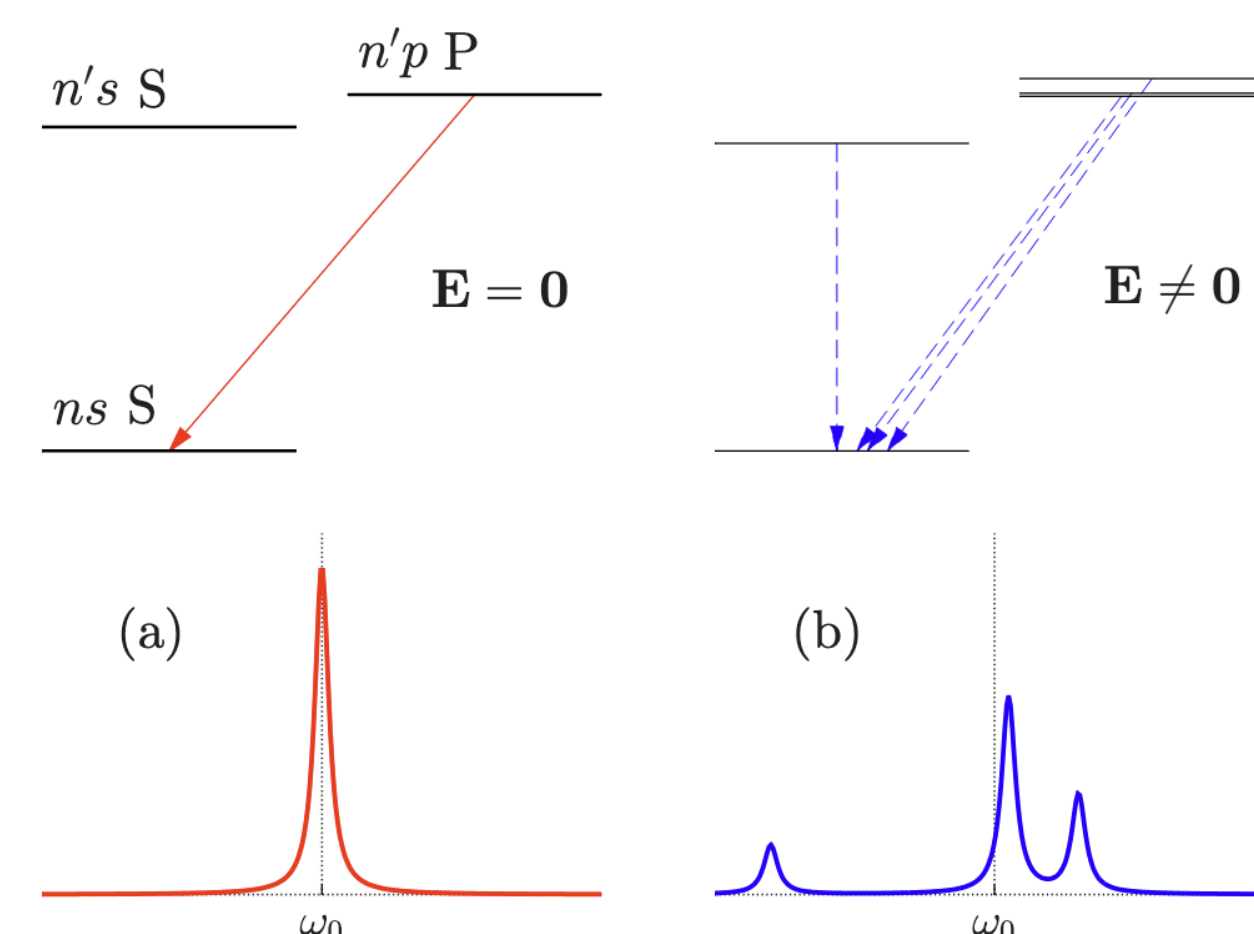


Fig. 2: Stark broadening in a stationary electric field. The left shows a regular transition with its corresponding opacity line, but the P and S states become coupled due to the presence of an electric field on the right, affecting the line shape. [3]

- Existing opacity models rely on outdated, semi-empirical methods to calculate line shapes. [2]

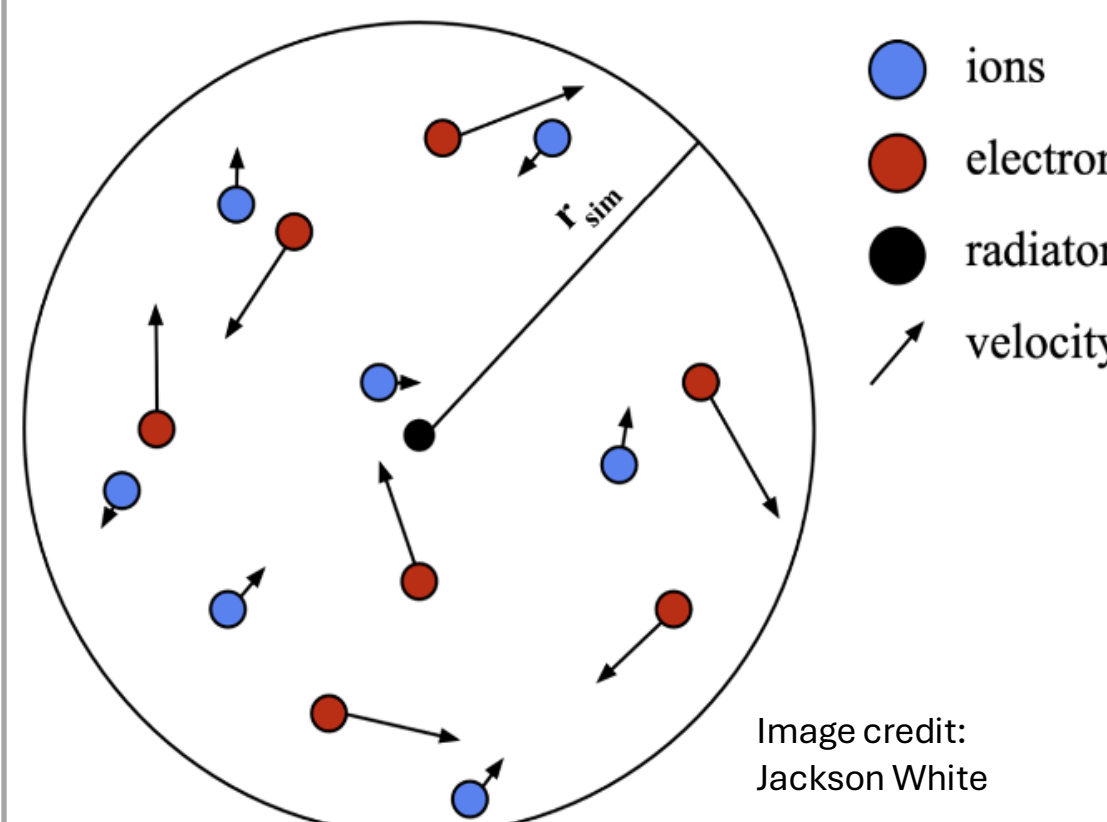


Fig. 3: An example of the dynamical model used in the line broadening program Xenomorph [4] (Unrelated to Fig. 4). Xenomorph uses straight-line particle trajectories, which are suitable for weakly coupled, low-density plasmas but break down in the HED regime. However, incorporating physically-motivated dynamics could improve the resulting line shapes.

- A discrepancy appears in iron plasmas at high temperature and relatively high density, requiring further investigation into its source.

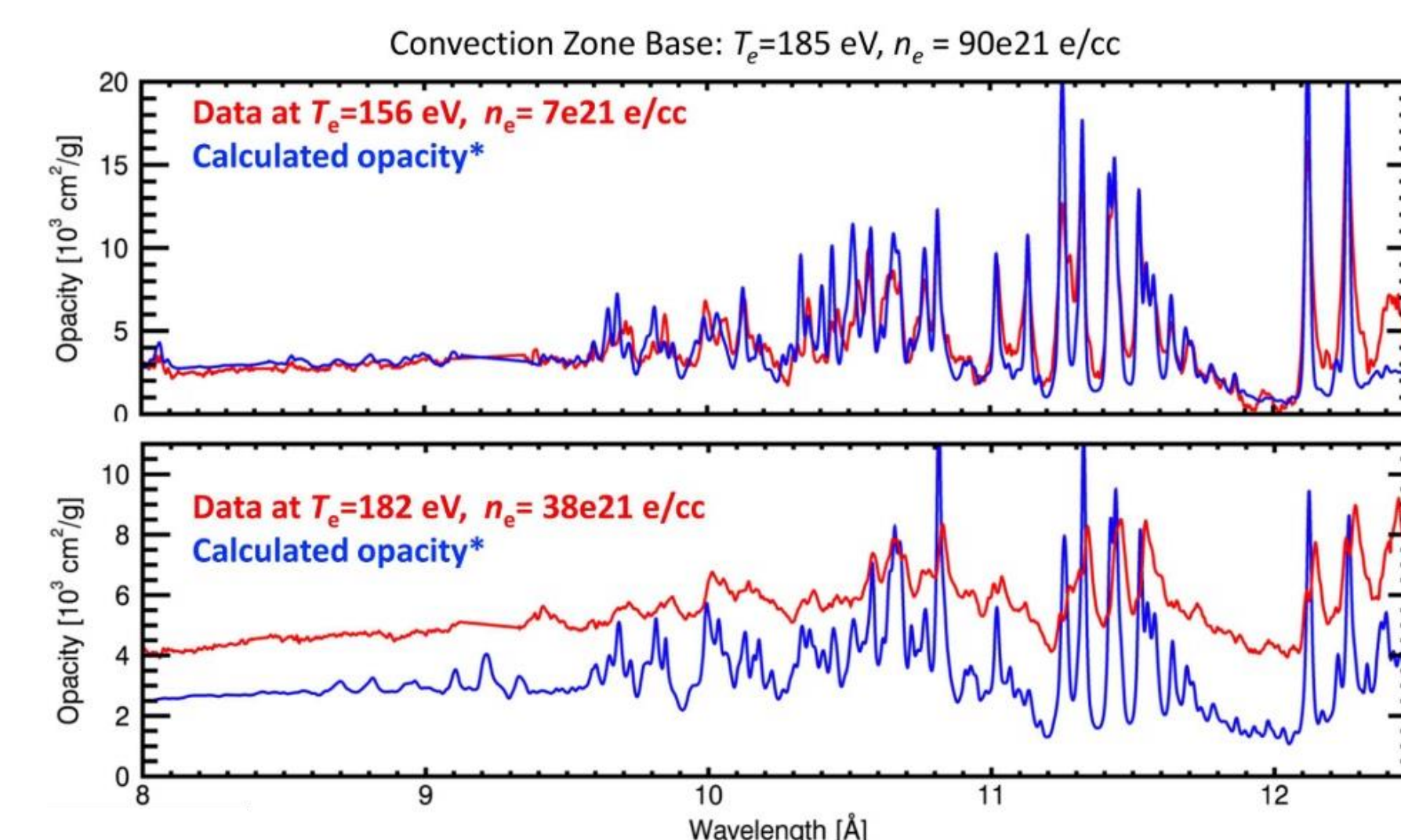


Fig. 4: Iron plasma opacity data (red) and calculations (blue) at two electron number densities and temperatures. At higher temperature and density (bottom) there is a discrepancy between the observed and modeled line shapes of both the bound-bound (horizontal lines) and bound free (spectral lines) transitions. [1]

- We hope to explore the physics at play through simulations of the bound-free transitions in simpler hydrogen plasmas at various temperatures and densities to assess the fidelity of newer computational methods.

Molecular Dynamics (MD) for Microfields

- Plasmas have stochastic microfields, which causes Stark broadening.
- Line shape codes use the microfield histories of a plasma simulation to calculate its corresponding spectrum. This computational method, introduced initially by [5], follows the process depicted in Fig. 5.

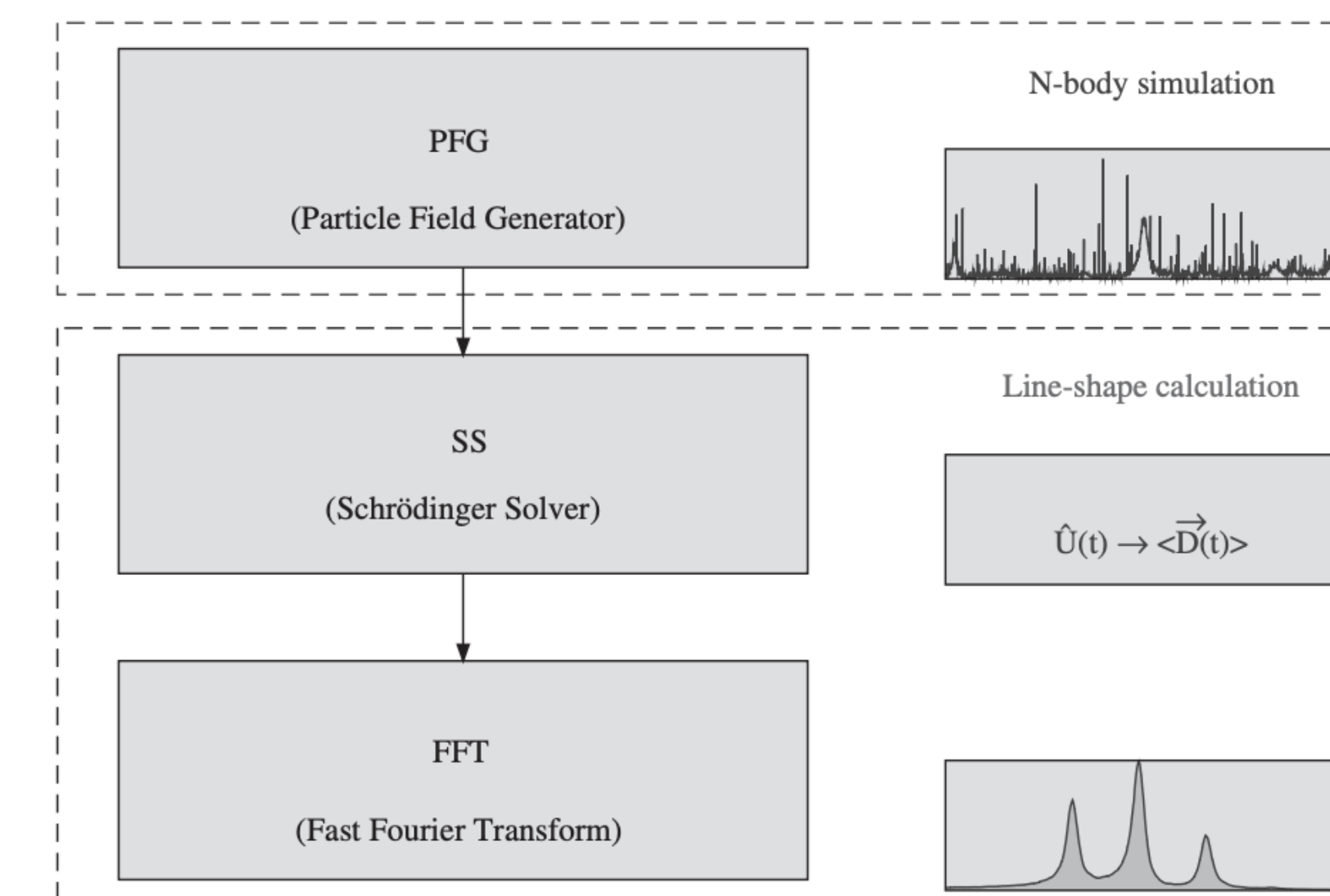


Fig. 5: A line shape simulation method presented by [5].

- Most line shape codes neglect interactions between plasma particles, but another option is to fully model the system via molecular dynamics (MD).
- Implementation of full MD has been shown to increase broadening in spectral lines as compared to the trivial, semi-empirical methods. Whether this means the model has more accurate line shapes is yet to be explored.
- Thus, physically-motivated molecular dynamics clearly impacts line shape calculations. We will conduct our study using full MD simulations to help determine whether this truly is a source of the spectral discrepancies.

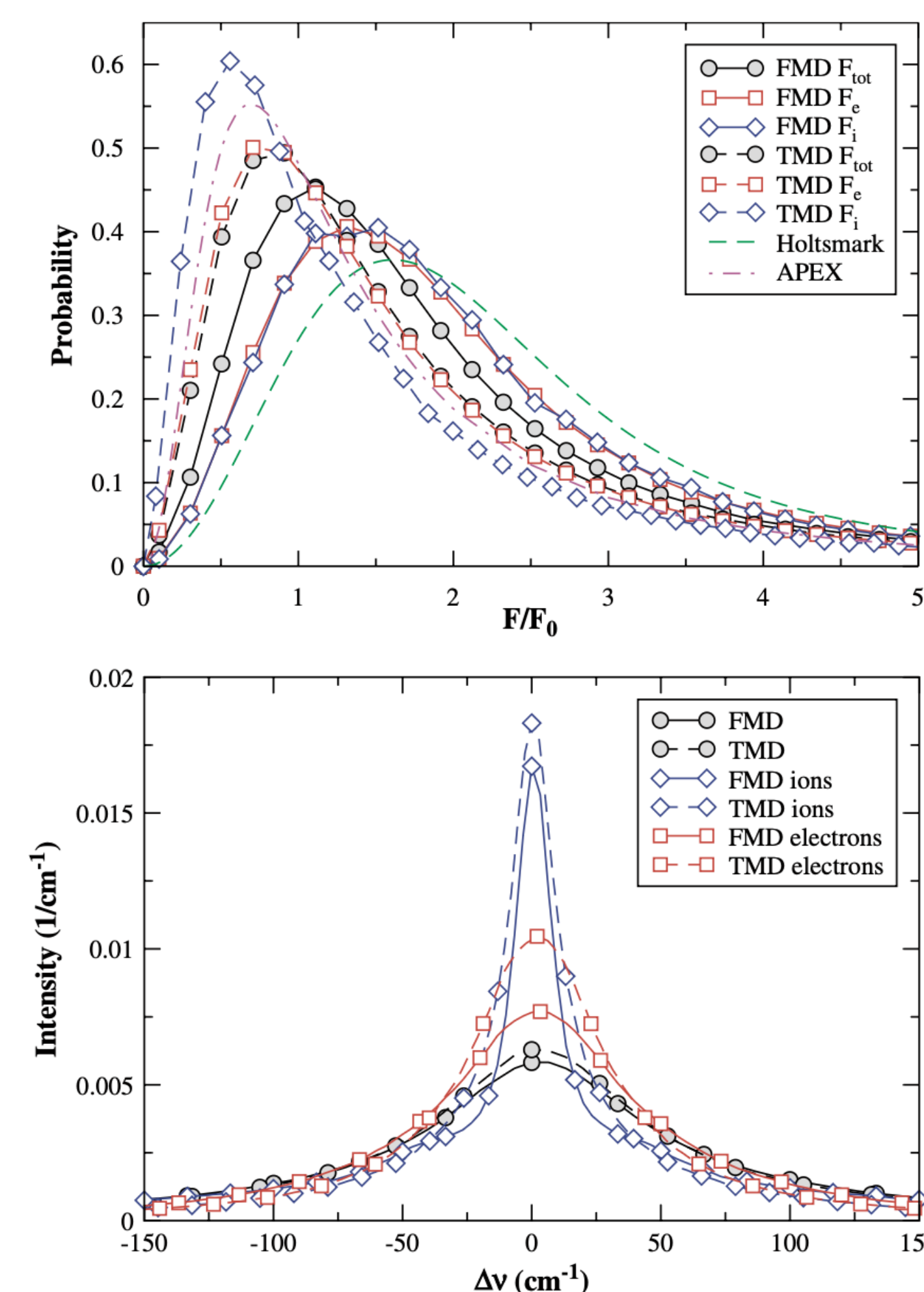


Fig. 6: Results from [6] showing the microfield distribution (top) and resulting line shapes (bottom) for a system of 200 hydrogen ions and 200 electrons at 1eV. Full MD (FMD) calculations produce broader lines than the trivial MD (TMD; straight-line trajectories) simulations.

- The conclusion of [6] claims that TMD is satisfactorily accurate; however, the existence of the iron opacity discrepancy in Fig. 4 shows that this is not true for higher temperatures and densities and with other ion species.

Preliminary Results

- I demonstrated that ddcMD can successfully produce microfield distributions that resemble those of [6].

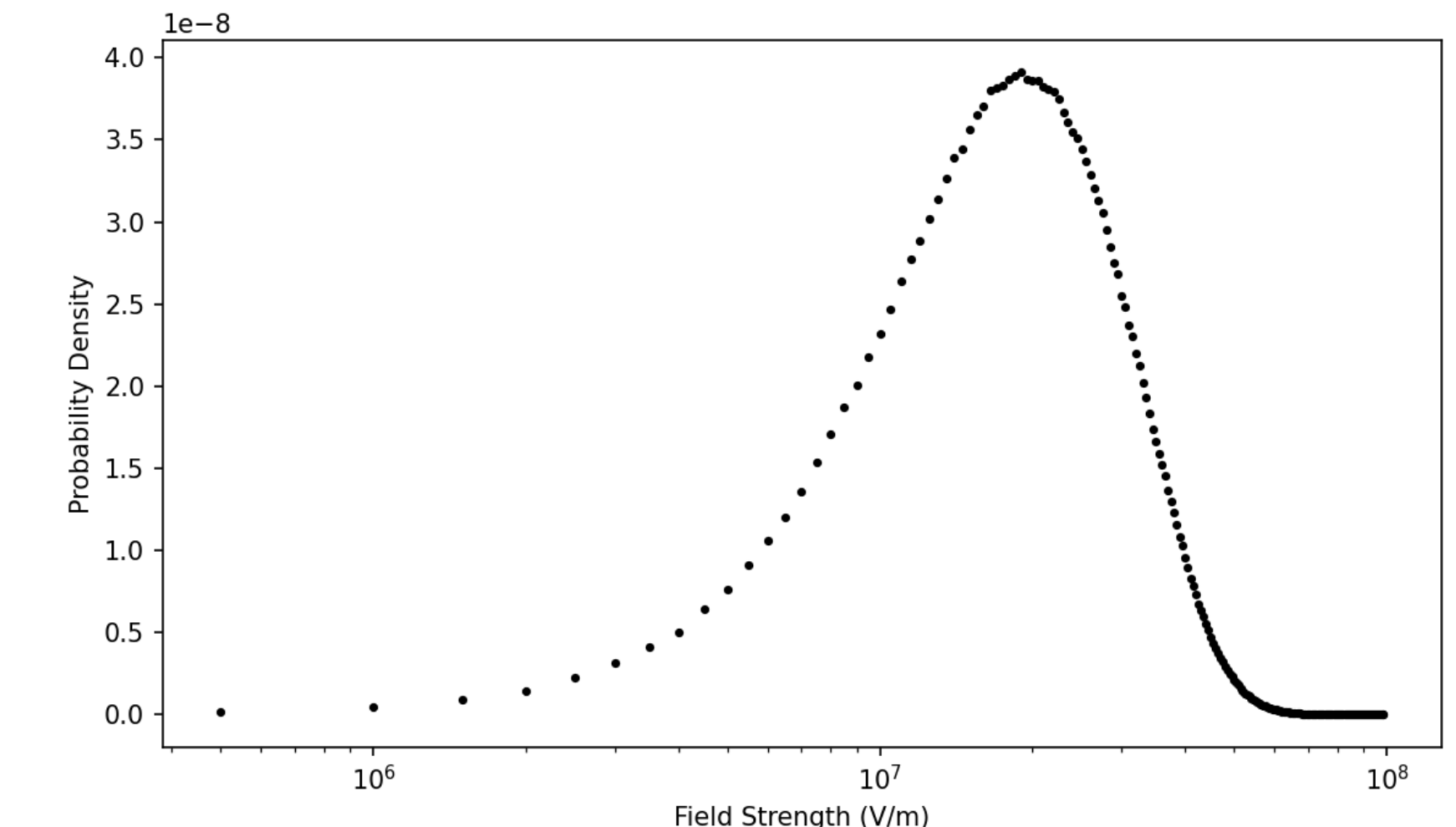


Fig. 7: My initial microfield distribution test for a hydrogen plasma of 128,000 particles at 1eV. The skew, as compared to the top panel of fig. 6, is simply a result of the logarithmic x-axis.

- I have also run the code for a system of deuterium atoms, taking inspiration from plasma conditions in [5].

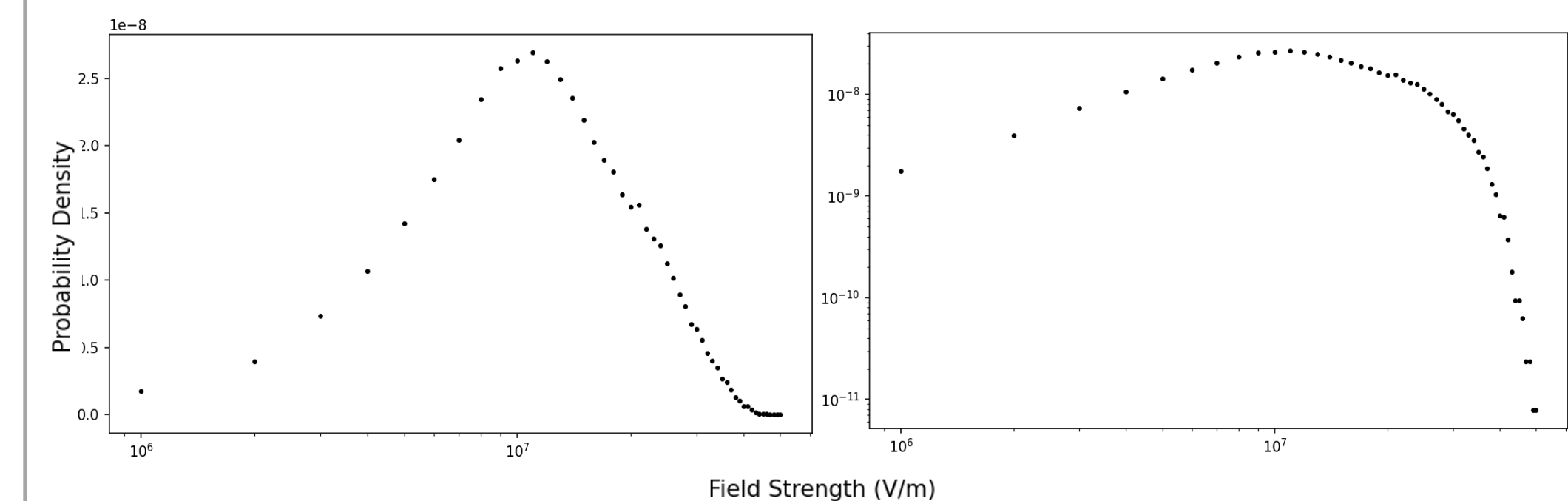


Fig. 8: A deuterium simulation plotted with a linear y-axis and logarithmic x-axis on the left, and both axes logarithmic on the right.

- The shape of the right side of the log-log plot in Fig. 8 should be less steep. This reflects an insufficient amount of rare events with large electric fields.
- From here, we will optimize computational parameters, starting with the Ewald parameter—which controls how real space and Fourier space calculations are split between tasks—and the size of the Fourier space.

Future Work

- Reproduce the simulations in Fig. 6 [5] and continue to higher temperatures.
- Simulate a variety of conditions to ensure that ddcMD has physically motivated line shape calculations.
- Optimize computational parameters (bold parameters will be studied this summer, the rest will be tested over this upcoming academic year).

- Ewald parameter α**
- Fourier grid size**
- Time step
- Total simulation length
- Simulation box size

References

- [1] Bailey et al. 2015 *Nature* 517 14048
- [2] Dimitrijevic & Konjevic 1980 *JQSRT* 24 451-459
- [3] Gigosos, Marco Antonio 2014 *J. Phys. D: Appl. Phys.* 47 343001
- [4] Gomez, Thomas 2013 *UT Austin, MS Thesis*
- [5] Stambulchik & Maron 2006 *JQSRT* 99 730-749
- [6] Stambulchik et al. 2007 *Science Direct: HEDP* 3 272-277